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Factors associated with cancer disclosure in adolescent and young adult cancer survivors: An integrative review from the social-ecological model perspective

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ABSTRACT

Purpose: Young cancer survivors ("young survivors") may need to disclose their cancer experiences to reintegrate into society. In such cases, the recognition of social support through the disclosure of cancer experiences may prevent potential social disadvantages .
This review aimed to describe the motivations, strategies and outcomes, and benefits and disadvantages of disclosure in young survivors based on the social-ecological model (SEM) to identify the support survivors need when disclosing their cancer experiences.
Methods: Using the integrated review methodology, we systematically searched six databases in English and Japanese as well as searched the reference lists of the selected studies. The themes identified via thematic analysis were categorized within the SEM levels.
Results: This review analyzed 14 studies and identified four themes, including "Motivation for Cancer Disclosure," "Barriers to Cancer Disclosure," "Consequences of Cancer Disclosure: Benefits," and "Consequences of Cancer Disclosure: Disadvantages." Motivations for young survivors to disclose their cancer involved post-cancer differences, perceptions, relationships, and social context. In navigating barriers, including self-stigma, peer exclusion, and discrimination, they employed strategies such as reassurance and information limitation. Tailored disclosure strategies at each SEM level offered social and psychological benefits, however, disadvantages, including stress, vulnerability, employment issues, and limited insurance coverage, were experienced by young survivors due to cancer disclosure.
Conclusions: To optimize the benefits of cancer disclosure for young survivors, addressing psychological burdens, enhancing disclosure skills, offering familial psychological support, and fostering public awareness of cancer are essential.

1. Introduction

A cancer survivor is defined as an individual who lives from diagnosis to the end of life (Marzorati et al., 2017). Notably, adolescents and young adults (AYA) affected by cancer, referred to as "young survivors" in this study, may experience long-term risks in cognitive, psychosocial, physical functioning, and economic statuses (Close et al., 2019; Hughes and Stark, 2018; Irene Su et al., 2018; Miller et al., 2020; Schilstra et al., 2021). For example, for young survivors, issues such as negative body image, perceptions that they are different from others, and cultural and social stigma toward cancer (Schilstra et al., 2021), as well as difficulties in maintaining jobs and personal relationships and reduced learning ability (Close et al., 2019) have been reported. In such a situation, young survivors develop autonomy and identity (Barnett et al., 2016; Cassano et al., 2008) and face the challenges of participating in a new society, such as studies, employment, and marriage. Young survivors with changes in physical functioning (such as decreased physical strength and visual function) and appearance (such as hair loss) due to cancer and treatment (Zebrack and Isaacson, 2012) may need to explain their cancer experience and its impact to those around them to gain

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understanding and support from friends, colleagues, romantic partners, and others as they resume social activities and continue or build new human relationships (Ke et al., 2020; Thompson et al., 2013). To identify the support for young survivors in a social context, this study discussed the factors and challenges associated with the disclosure of cancer experiences by young survivors in a social context.

Many adult cancer survivors recognize the advantages of cancer disclosure (Munro et al., 2015). There is some evidence that cancer disclosure helps adult cancer survivors gain social support, reduce fear, and gain a sense of security, which facilitates adjustment to treatment (Lepore et al., 1996; Munro et al., 2015; Zhang et al., 2019). However, not only cancer disclosure but disclosure per se is not always advantageous. Disclosure may cause risks and disadvantages. The risks include uncertainty and some types of loss or damage that may occur (Kaplan and Garrick, 1981). Subjective disclosure risks include loss of privacy and exclusion or rejection from people around them (Dienlin and Metzger, 2016; Krasnova et al., 2010). Disadvantages of cancer disclosure in adult survivors, including loss of self-control and autonomy, sympathy of others (Munro et al., 2015), tension in personal relationships, and avoidance of others, have also been reported (Else-Quest and Jackson, 2014; Fujisawa et al., 2020; Yoo et al., 2010). A review of couples' communication about cancer (Hasson-Ohayon et al., 2022) suggested that self-disclosure by the survivor and his or her partner provides advantages regarding patient adjustment to treatment and reduced psychological suffering. When couples avoid talking about cancer or feel socially constrained, the psychological well-being of the survivor and his or her partner decreases (Butner-Kozimor and Savla, 2020; Lyons et al., 2016; Manne et al., 2015; Wertheim et al., 2018). On the other hand, discrepancies in self-disclosure within a couple pose a risk of a detrimental relationship, indicating that disclosure between a survivor and his or her partner is not always advantageous (Hagedoorn et al., 2000).

Cancer disclosures of young survivors with different developmental challenges from adults, may differ from those of adult survivors. Young survivors (ages 20-39) have been reported to have significantly higher psychological suffering prior to disclosure than adult survivors (ages 40-59) (Tsuchiya et al., 2022). In addition, fear of a negative impact of the disclosure on personal relationships and poor peer response to the disclosure are barriers to communication about cancer in young survivors (Janin et al., 2018). These findings highlight the large barriers to cancer disclosure for young survivors. However, very few studies of cancer disclosure have been limited to the younger generation. Therefore, experiences of cancer disclosure by young survivors with unique developmental challenges are not well known. Since the disadvantages of cancer disclosure in young survivors can create fear of future disclosure, exploring the advantages, disadvantages, and barriers to self-disclosure in young survivors, would provide resources to help them gain the advantages of disclosure while minimizing the disadvantages.

Cancer disclosure affects not only the cancer survivor but also others to whom the disclosure is made and society. The response of the disclosed recipient can be to the benefit or disadvantage of the cancer survivor and sometimes a barrier to future disclosure. Interactions by the survivor and disclosed recipients, such as the impact of cancer disclosure communication on the psychological well-being of the young survivor and spouse, can be to the benefit or detriment of both parties. Therefore, conceptualizing cancer disclosure as an interaction between individual and social-environmental factors would facilitate understanding of cancer disclosure among young survivors. Notably, the social-ecological model (SEM) (Eddy, 1981; McLeroy et al., 1988) is a conceptual framework for identifying factors that influence interactions and outcomes in social-ecological systems (Partelow, 2018). SEM serves as an appropriate framework in the context of cancer survivorship, providing a foundation for designing and implementing interventions aimed at improving the quality of life for survivors (Moore et al., 2015). Already, studies on the sexual identity of lesbian, gay, bisexual, and transgender people; human immunodeficiency virus; and sexually transmitted disease disclosure (Emetu and Rivera, 2018; Qiao et al., 2015; Restall et al., 2021) support the application of a social ecological approach in the disclosure analysis of stigmatized individuals.

This review aims to identify the necessary support for young cancer survivors when disclosing their cancer experiences, considering the benefits and disadvantages of disclosure. First, we will use SEM to identify the factors that contribute to cancer disclosure among young survivors at each level, and we will describe the interactions among these factors within and across multiple levels. Secondly, we will review the support needs based on the benefits and disadvantages of cancer disclosure. We will also propose recommendations for the necessary support. To the best of our knowledge, there are no reviews of cancer disclosures by young survivors from a social-ecological perspective. This review will take a social-ecological perspective to encourage the development of support programs for young survivors' cancer disclosures and contribute to their reintegration into society.

2. Methods

2.1. Design

We planned to conduct an integrative review for assessing the literature (Hopia et al., 2016; Whittemore and Knafl, 2005). An integrative review can include various types of study designs, such as qualitative and quantitative analyses.

2.2. Data sources and search strategies

We performed an electronic search for articles related to our objective. CINAHL, PubMed, PsycINFO, and Web of Science databases were searched for English articles related to medicine and psychology, and Igaku Chuo Zasshi and Cinii databases were searched for Japanese articles using the following search terms: ("pediatric" OR "child" OR "adolescent" OR "AYA") AND ("cancer" OR "oncology" OR "neoplasm" OR "malignancy" OR "tumor" OR "leukemia" OR "lymphoma") AND ("disclosure" OR "self-disclosure.") For searching Japanese articles, "childhood cancer," "experience," and "nursing" were added to the search terms (see Supplementary Material 1). A reference list of selected references was reviewed to supplement the literature search. The references ranged from the years 2000 (When psychosocial issues among AYA cancer patients began to attract attention, e.g. (Bleyer, 2007)) to 2021 were searched for May 2021. Initially, NH conducted the search, identification, screening, eligibility, and incorporation, which were subsequently verified by IS.

2.3. Inclusion and exclusion criteria for screening literature

The inclusion criteria were as follows: a) studies exploring the effect of disease disclosure on the social lives of AYA cancer survivors; b) studies with patients aged 10–39 years at the time of the inclusion (WHO definition of adolescence [World Health Organization. Adolescent health.], 10–19 years; AYA criteria (Desandes and Stark, 2016; Salloustros et al., 2017), 15–39 years), including those that had "adolescent" or "young adult" in the title when age at diagnosis was unknown; c) qualitative, quantitative, mixed-method, or review studies; d) studies published in peer-reviewed journals; and e) studies published in English or Japanese. Studies of participants other than cancer survivors, theses on theory and scale development, and graduation theses were excluded from the review.

2.4. Quality appraisal

Two reviewers independently used the Johns Hopkins Nursing Evidence-Based Practice Rating Scale (Dang and Dearholt, 2017) to evaluate the quality and robustness of the studies that met the inclusion criteria. The robustness of evidence was evaluated as follows: Level I:

experimental study, randomized controlled trial, or meta-analysis; Level II: a quasi-experimental study; and Level III: nonexperimental study, qualitative study, or meta-synthesis. The quality of evidence was classified as high (A), good (B), or low/with severe defects (C). Several studies were discussed in person to obtain a consensus on the level and grade of evidence. The first [NH] and second [YS] reviewers evaluated the data. A third reviewer [AM] participated as needed to discuss and compare outcomes, resolve disagreements, and reach a consensus. The 19 studies included in the final review were all level III; of which, 6 studies were graded A, 8 were graded B, and 5 were graded C.

2.5. Analysis of the extracted data

For data analysis, all the articles extracted based on the inclusion and exclusion criteria were read to get an overall picture by all authors. Subsequently, the two authors (NH and IS) independently read the papers again, and data was extracted from the selected papers using the matrix method (Garrard, 2013). Disagreements among the researchers were discussed and a consensus was reached. Study author(s), year, country, objective, design, sample size/age range/sex, SEM level, and evidence level were included in this matrix.

Thematic analysis (Clarke and Braun, 2014) with an inductive approach was used to analyze and synthesize the evidence from NH's extracted data (results section) and IS confirmed. Thematic analysis is a flexible approach in which philosophical standing is not an issue and can be used in qualitative research alone or mixed research methods (Clarke and Braun, 2014). Themes were generated by NH and IS. In addition, themes were categorized into the appropriate levels (i.e., intrapersonal, interpersonal, institutional, community, and policy levels) using the framework of SEM.

3. Results

A comprehensive search generated 775 results; of these, 328 duplicate results were deleted (Fig. 1). The remaining 447 results were screened by title, abstract, and exclusion criteria; 415 more results were eliminated consequently. After conducting a complete review based on the inclusion and exclusion criteria, 15 more results were eliminated. We reviewed the 17 articles on the reference list that we found to be accurate and added 2 more, leading to a total of 19 references. Articles

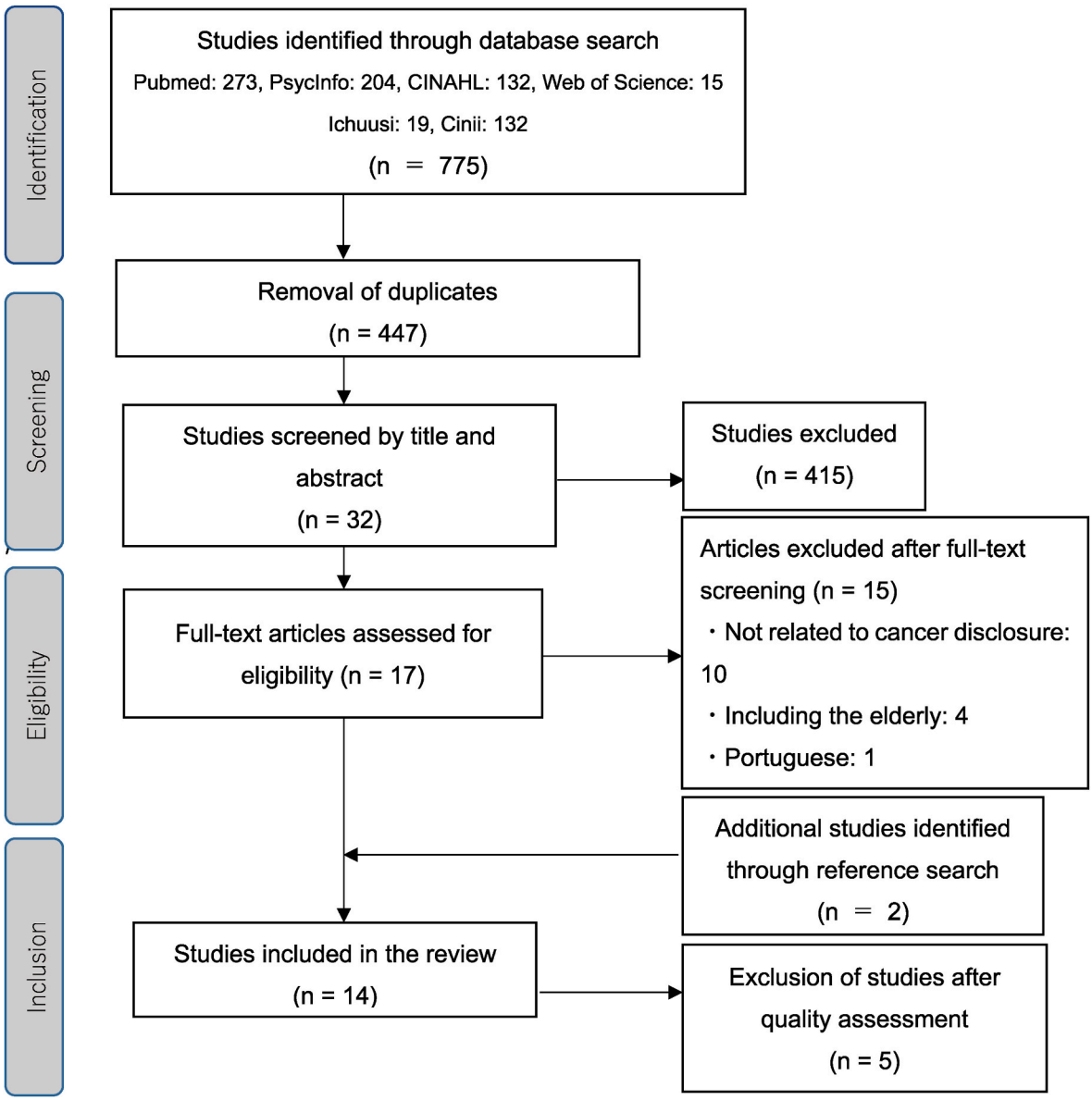

Fig. 1. Literature search strategy and flow.

evaluated as grade C during quality assessment were further eliminated, leaving 14 articles that were ultimately analyzed in this review. Table 1 summarizes the articles and their level of evidence. The papers to be reviewed included two systematic reviews of qualitative studies, six qualitative studies, five quantitative studies, and one mixed-method study. Studies conducted in the US, Korea, Australia, Belgium, Canada, and the UK accounted for five, three, two, two, one, and one studies, respectively. Five studies included participants aged ≥40 years (the age distribution of participants, calculated from mean age and SD, was 70%–95% for patients aged <40 years). The review included participants of all ages, ranging from teenagers to individuals in their 40s, who had been diagnosed with cancer at different times. As a result, the study reported on the experiences of young cancer survivors in disclosing their diagnosis at various stages of life (school, work, etc.) and to different recipients (friends, colleagues, romantic partners, spouses, and family members). It was found that young survivors, whose life stages were rapidly changing, required cancer disclosures tailored to their specific stage of life and the individuals they were disclosing to.

All studies were conducted in or after 2011. Ten studies focused on disease disclosure and communication; one of them also focused on advocacy. The Social Constraints Scale (Kim and Yi, 2014; Lepore et al., 1996), which was used as a self-disclosure scale for stressful events, the Nebraska Outness Scale (Meidlinger and Hope, 2014; Rabin, 2019), which was used as a sexual minority self-disclosure scale, or the self-made Likert scale were used in the quantitative studies.

A thematic analysis of the 14 articles revealed four main themes:

Table 1
Summary of studies.

Authors (year) Country	Study aim	Design	Sample size/age range/sex	Levels of SEM	Levels of evidence
Belpame et al. (2019) Belgium	To seek insights into how a childhood cancer experience plays a role in the daily life of adolescents and young adult (AYA) survivors.	Qualitative	N = 21/14–25 years/females = 13, males = 8	Intrapersonal Interpersonal	III-A
Carpentier et al. (2011) USA	To examine AYA testicular cancer survivors' subjective understanding of the impact of cancer in adolescence and young adulthood, with a particular emphasis on romantic and sexual relationships.	Qualitative	N = 21/18–34 years/males = 21	Intrapersonal	III-B
Darabos (2019) USA	To explore the moderating role of the communication mode on the relationship of relational processes of disclosure and responsiveness with psychological well-being.	Quantitative: Cross-sectional study	N = 62/24–42 years/females = 59, males = 3	Intrapersonal	III-B
Dobinson et al. (2016) Australia	To delineate the pathways through which AYA survivors' unmet psychosexual needs manifest and demonstrate further the types of unmet psychosexual needs experienced by AYAs.	Qualitative: Grounded theory methodology	N = 11/15–45 years at the time of diagnosis/females = 6, males = 5	Intrapersonal	III-A
Easley (2019) Canada	To gain an in-depth understanding of the motivations for cancer history disclosure and/or nondisclosure among young adult cancer survivors.	Qualitative: Constructivist grounded theory approach	N = 28/19–38 years at the time of diagnosis, 21–52 years at time of the interview, females = 16, males = 12	Intrapersonal Interpersonal	III-A
Janin et al. (2018) Australia	To analyze how childhood cancer patients and survivors communicate information about their cancer with family and peers.	Systematic Reviews	N = 6 qualitative studies/16–34 years/number of participants and sex unknown	Intrapersonal Interpersonal Community	III-B
Kim and Yi (2014) Korea	To examine how perceived public stigma affects psychological distress as mediated by cancer disclosure, internalized reactions to stigma, and social support availability.	Quantitative: Cross-sectional study	N = 223/15–39 years/females = 92, males = 131	Intrapersonal	III-B
Kim et al. (2018) Korea	To identify young adult Korean cancer survivors' individual (psychological distress, stigma, sociodemographic variables, and cancer-related variables) and network-level factors (relationship type, social support type) that influence the discussion of their cancer experiences.	Quantitative: Cross-sectional study	N = 68/19–33 years/females = 34, males = 34	Intrapersonal Interpersonal	III-B
Lahaye et al. (2017) Belgium	To identify and evaluate 1) the challenges and difficulties that AYA survivors of hematopoietic stem cell transplantation (HSCT) still experience due to their past illness and its complications and 2) strategies or resources of AYA survivors of HSCT who exploit to help them in dealing with the current implications of their past medical experiences.	Qualitative	N = 30/23.61 (SD: 5.21) years/ females = 12, males = 18,	Intrapersonal Interpersonal	III-A
Pini et al. (2019) UK	To explore the perspectives of young people as they return to school during and after curative cancer treatment.	Qualitative: Longitudinal study	N = 12/13–16 years/females = 6, males = 6	Interpersonal Organizational	III-A
Rabin (2019) USA	To shed light on the processes by which young adult cancer survivors self-disclose their cancer history and experiences to peers.	Quantitative: Cross-sectional study	N = 122/19–45 years/females = 98, males = 24	Intrapersonal Interpersonal Community	III-B
Rabin (2020) USA	To evaluate the factors impacting AYA survivors' decision of disclosure in the workplace/school and strategies used for self-disclosure.	Quantitative: Cross-sectional study	N = 120/15–39 years at the time of diagnosis, 19–45 years at the time of the interview/females = 96, males = 24	Intrapersonal Interpersonal	III-B
Wong et al. (2017) USA	To provide a phenomenological synthesis of high-quality papers found in the published literature related to AYAs' unmet needs following primary treatment for cancer.	Systematic review	N = 58 studies/number of participants = 1993/16–39 years/sex unknown	Intrapersonal Interpersonal	III-A
Yi and Nam (2017) Korea	To investigate both strategies for coping with stigma and the effects of participating in advocacy.	An embedded-design mixed-method study	Interview N = 30/18–35 years/females = 14, males = 16 Survey N = 145/20.3 (SD: 3.67) years/ females = 69, males = 76	Intrapersonal Interpersonal Organizational Community	III-B

"Motivations for Cancer Disclosure," "Barriers to Cancer Disclosure," and "Consequences of Cancer Disclosure: Benefits," and "Consequences of Cancer Disclosure: Disadvantages." Further, each theme was categorized into intrapersonal, interpersonal, institutional, and community levels using a SEM framework. This review did not identify any policy-level factors. Themes and factors related to adolescent to young adult cancer disclosure, analyzed using a SEM, are presented in Table 2. Additionally, a summary of the identified themes and representative quotes is provided in Table 3.

3.1. Intrapersonal level

The intrapersonal levels in the SEM (McLeroy et al., 1988) are an individual's beliefs, knowledge, attitudes, behaviors, and self-concept.

3.1.1. Motivation for cancer disclosure

At the intrapersonal level, factors that motivated cancer disclosure included changes in appearance and behavior before and after the young survivor's cancer experience (appearance and behavior), as well as perceptions of the cancer experience (important factors that shape who we are, the need to understand the cancer experience, and altruistic motives as advocates). The factors that most frequently motivated disclosure were differences in their appearance after their cancer experience, certain things young survivors could not do, and the need to explain their particular behaviors (Belpame et al., 2019; Easley, 2019; Janin et al., 2018; Lahaye et al., 2017). Young survivors expected that they should to be treated as normal while considering their cancer experience (Belpame et al., 2019). Young survivors further expressed a need to talk to others about their past based on the realization that their cancer experiences were important to their lives and a defining element of their personality (Belpame et al., 2019; Lahaye et al., 2017). Some participants talked to their parents about their illness because of it out of a need to understand the cancer experience (Lahaye et al., 2017). The motivations of young survivors to participate in advocacy activities included supporting other patients and families (Belpame et al., 2019; Easley, 2019; Rabin, 2019) and promoting understanding regarding cancer experiences in the general population (Rabin, 2019; Yi and Nam, 2017).

3.1.2. Barriers to cancer disclosure: exclusion from peer groups and society as well as self-stigma

The barriers to cancer disclosure were classified into two categories: rejection risk by peers and society and self-stigma. Young survivors feared discrimination and prejudice from friends, peers, or someone with whom they have an intimate relationship, as well as rejection from the group, and they were worried about being viewed as weak people and being the object of excessive pity or compassion (Easley, 2019; Janin et al., 2018; Lahaye et al., 2017; Rabin, 2020; Yi and Nam, 2017). Survivors with a romantic partner or spouse harbored concerns regarding the timing of disclosure (Carpentier et al., 2011), sexual problems, and disclosure of emotional aspects of cancer (Darabos et al., 2019; Dobinson et al., 2016; Janin et al., 2018). Young survivors with internalized shame, psychological distress, and greater self-stigma toward cancer were significantly less likely to disclose cancer (quantitative studies (Kim and Yi, 2014; Kim et al., 2018; Yi and Nam, 2017),).

Table 3
Representative quotes of the theme at each SEM level.

Intrapersonal level	
Motivation for Cancer Disclosure	"But I felt like I wanted them to know … I need people to realize that I'm not this strange weird introverted person." (Easley, 2019)
Barriers to Cancer Disclosure	"… the whole opening up thing and telling someone, it's really hard because I don't want a pity party …" (Wong et al., 2017)
Interpersonal level	
Motivation for Cancer Disclosure	Participants sometimes felt "bombarded" with this flurry of questions. (Pini et al., 2019)
Barriers to Cancer Disclosure	"I had some people I was dating who reacted negatively to the scars and some who just play the sympathy card …" (Wong et al., 2017)
Consequences of Cancer Disclosure: Benefits	"… It is how I felt about it and … because if everybody else thinks that I am going to beat it then they are all going to support me to help me beat it, …" (Pini et al., 2019)
Consequences of Cancer Disclosure: Disadvantages	" …, Like when I had the operation just on the massive lump I had, …, everybody asked me, what's that, what is it and I didn't know so I just said, I don't know but everybody asked me it, virtually everybody and it just got really irritating and annoying, you know." (Pini et al., 2019)
Institutional level	
Barriers to Cancer Disclosure	"… one of my friends does not go to any hospital so he can join private insurance … He tried to find another way, but this is the only possible way." (Yi and Nam, 2017)
Community level	
Consequences of Cancer Disclosure: Benefits	"A feeling that I can do something? Accomplishment? Participatimg in those activities, I learned how to meet and treat people …" (Yi and Nam, 2017)

3.2. Interpersonal level

The interpersonal factors in the SEM (McLeroy et al., 1988) are interpersonal processes and primary groups. Interpersonal are formal and informal social support systems, including family, friends,

Table 2
Themes and factors in adolescent to young adult cancer disclosure using a social–ecological model.

Themes	Motivation for cancer disclosure	Barriers to cancer disclosure	Consequences of cancer disclosure: Benefits	Consequences of cancer disclosure: Disadvantage
Levels of SEM			Factors	
Intrapersonal	✓ Differences from before cancer experience ✓ Recognition of Cancer Experience	✓ Exclusion from Peer Groups ✓ Self-Stigma		
Interpersonal	✓ Socio-cultural context ✓ Relationships with others	✓ Reaction of the disclosure recipient ✓ Lack of understanding	✓ Disclosure strategy: reassurance, humor ✓ Social support ✓ Psychological benefits	✓ Disclosure strategy: information limitation ✓ Lack of cancer knowledge among young survivors ✓ People's lack of understanding of young cancer survivors ✓ Hurt family members
Institutional		✓ Discrimination	✓ Employment strategies through cancer disclosure	✓ Employment strategies through cancer disclosure
Community			Benefits ✓ Online cancer disclosure ✓ Advocacy participation	

schoolmates, and workgroups.

3.2.1. Motivation for cancer disclosure

Situational factors, such as social and cultural background and relevance to a topic of conversation or timing, all influenced their decision to disclose, regardless of their motive for disclosure (Easley, 2019). For example, it is rare in a school for a student to get cancer; thus, young survivors often disclosed information in response to their peers' and faculty members' request (Belpame et al., 2019; Lahaye et al., 2017; Pini et al., 2019). Relationship with the others was a factor in their cancer disclosure decision (Easley, 2019; Janin et al., 2018). Survivors tended to speak about their cancer experience with family, intimate partners, friends, and people with whom they were in a reciprocal support relationship (Kim et al., 2018; Rabin, 2020).

3.2.2. Barriers to cancer disclosure: reaction of the disclosure recipient and lack of understanding

Young survivors described their experiences of resistance to cancer stories, curiosity about cancer, and lack of understanding demonstrated by their disclosure partners as barriers to cancer disclosure (Janin et al., 2018; Wong et al., 2017). The satisfaction of young survivors with the response of the person to whom they disclosed cancer was associated with the likelihood of future disclosure [quantitative studies, (Rabin, 2019, 2020).]

3.2.3. Consequences of cancer disclosure: benefits

The benefits of cancer disclosure for young survivors include enhanced empowerment, sense of control, social support, and reduction in psychological distress. Disclosure strategies that focused on positive responses and support from the disclosure recipient helped young survivors gain a sense of empowerment, control (Easley, 2019), and social support from the disclosure recipient. The most common disclosure strategy used was to reassure others, use humor, and show that they are okay (Janin et al., 2018; Lahaye et al., 2017; Rabin, 2019, 2020). Young survivors who made their friends feel more secure during their disclosure rated their friends' responses as more positive than survivors who did not [Quantitative study $t = 2.0$, $p = 0.04$ (Rabin, 2019),]. Moreover, some young survivors shared their cancer stories in positive ways because they believed that it would be easier for their friends to support them if they felt that the survivors were adequately handling their disease (Pini et al., 2019). The benefits of disclosing cancer-related experiences included the psychological rewards of disclosure. For young survivors, communicating about their cancer-related experiences with others was important (Wong et al., 2017). Furthermore, sharing the psychological burden of cancer-related experiences with others formed a coping mechanism for the young survivors (Janin et al., 2018). In addition, cancer disclosure significantly improved psychological distress in young survivors by reducing internal shame [quantitative study $B = 0.186$, 95%CI = 0.920–0.998 (Kim and Yi, 2014),].

3.2.4. Consequences of cancer disclosure: disadvantages

The disadvantages of cancer disclosure for young survivors included feelings of frustration and increased vulnerability. Most young survivors limited the amount of information they shared with others to avoid drawing attention to themselves due to differences between them, as cancer survivors, and their peers (Belpame et al., 2019; Rabin, 2019). However, young survivors' ratings of less positive responses from friends were associated with limited information about disclosure [quantitative study, $t = 3.98$, $p = 0.000$ (Rabin, 2019),]. In addition, excessive emotions or incorrect information about their cancer experience shared with peers with whom they were not very close were a source of stress for young survivors (Pini et al., 2019). Moreover, the inability to adequately answer questions owing to the limited knowledge of the cancer experience led to frustration and vulnerability among the teenage survivors (Pini et al., 2019).

They also suppressed their emotions as a strategy to protect the feelings of their loved ones (Janin et al., 2018). While it was deemed necessary to speak about their past disease to close family and friends in some situations, in others, it became a taboo subject and was avoided to prevent hurting the family by discussing the survivors' past illness (Lahaye et al., 2017).

3.3. Institutional level

Institutional factors in SEM (McLeroy et al., 1988) include social institutions with organizational characteristics and operating rules.

3.3.1. Barriers to cancer disclosure: discrimination

Young survivors who faced discrimination at their schools and communities kept their cancer history a secret to avoid discrimination. Furthermore, they concealed their cancer history and stopped going for regular checkups to obtain private insurance (Yi and Nam, 2017).

3.3.2. Consequences of cancer disclosure: benefits and disadvantages

The disclosure of cancer by young survivors has generated both advantages and disadvantages, encompassing employment successes and failures as well as experiences of discrimination. Educational and medical professionals supported young survivors in disclosing cancer at school (Pini et al., 2019). However, they were repeatedly denied employment due to cancer history disclosure in job interviews (Yi and Nam, 2017). In contrast, some survivors successfully obtained employment after disclosing their cancer history and advocacy activities as overcoming experiences in interviews (Yi and Nam, 2017).

3.4. Community level

Community factors in the SEM model (McLeroy et al., 1988) are organizations, institutions, and informal networks within defined boundaries.

3.4.1. Consequences of cancer disclosure: benefits

For the young survivors, cancer information disclosure at the local level was a source of support, accomplishment, and social skills. Young survivors disclosed their cancer history online to share the information with a wider population. Blogs, patient- and family-only online communities (Janin et al., 2018), emails, and Facebook were the media used for this purpose (Rabin, 2019). Online platform was a place for survivors to freely vent their inner feelings, including medical and psychological issues (Janin et al., 2018; Rabin, 2019). Moreover, it was an important source of support for them, and online communication helped them cope with uncertainty as cancer survivors (Janin et al., 2018). In addition, participating in advocacy helped them increase their self-esteem and gain a sense of accomplishment and interpersonal social skills (Yi and Nam, 2017).

4. Discussion

Fourteen studies of cancer disclosure among young survivors were organized into four themes and four levels (individual, interpersonal, institutional, and community) using an SEM framework. Results identified 20 factors contributing to cancer disclosure among young survivors at all levels. Cancer disclosure was sometimes beneficial and detrimental for young survivors, depending on their disclosure motives and strategies. Quantitative studies concurred with or reinforced the results of qualitative studies.

4.1. Barriers to cancer disclosure

Barriers to disclosure at the intrapersonal level included fears related to discrimination and exclusion from others, internalized shame and psychological distress, and self-stigma among survivors. Barriers to disclosure at the intrapersonal level in young survivors were similar to

those in adult cancer survivors (Adams et al., 2015; Grunfeld and Cooper, 2012; Munro et al., 2015; Stergiou-Kita et al., 2016; Tsuchiya et al., 2015, 2022). However, to the authors' knowledge, "exclusion from others" is a barrier unique to young survivors, suggesting a stronger risk of disclosure for young survivors. Social interaction with peers is an important aspect of identity in young adults. Exclusion from others" can undermine the emotional well-being of young survivors through social interactions (Cassano et al., 2008).

The negative reactions of disclosing recipients (interpersonal) regarding being discriminated can become survivors' social constraints to disclosure in school or elsewhere. Young survivors' experiences of disadvantage can evoke fear of discrimination and exclusion, leading to internalized shame, psychological distress, and self-stigma at the individual level. These factors contribute to an increase in barriers to disclosure. The psychological barriers among young survivors and disclosure disadvantages that aroused at the interpersonal (e.g., disclosure recipient response) or organizational level were related to each other. The social cognitive processing model proposes the social constraints on cancer disclosure: Social constraints on disclosure are both objective social conditions and individual interpretations of those conditions that cause individuals to refrain from or change their disclosure of stress- and trauma-related thoughts, feelings, or issues. In addition, individual psychological distress affects social constraints (Lepore and Revenson, 2007). Hence, the individual-level psychological distress of young survivors may influence their perception of social constraints in cancer disclosure, such as interpersonal-level disclosing recipients' reactions and organizational-level discrimination (Lepore, 2001). It may be related to future disclosure intentions. Therefore, interventions for survivors' psychological distress are an important element of a program to support cancer disclosure for young survivors. Approaches to the public, such as public campaigns, are important to promote understanding of the cancer experience among the public. Better understanding by the public is expected to mitigate the social constraints of cancer disclosure.

4.2. Motivations for cancer disclosure and disclosure strategies and benefits of cancer disclosure

The young survivors used the understanding and support of others as their disclosure objective and used disclosure strategies, such as reassuring their recipients, to achieve that objective. As a result, they benefited from their sense of empowerment and control and social support from the disclosing recipients. This finding is consistent with the disclosure process model, which explains that approach-focused disclosure goals lead to adopting effective strategies and enable disclosure benefits (Chaudoir and Fisher, 2010). The psychological benefits of young survivors from advocacy activities are explained by the Disclosure Decision-Making Mode (DDMM) (Greene et al., 2012), which shows that disclosure motives that focus on the benefits of others become role models for others who are afraid to disclose and the psychological benefits from disclosure by recognizing the possibility of contributing to the greater good of society. That is, strategic cancer disclosure based on clear motivations has the potential to curb barriers to disclosure and capture the benefits of disclosure.

Furthermore, online disclosure helps survivors gain support from other survivors with the same experience, improve their social skills, and reduce their isolation (Bergeron et al., 2019; Perales et al., 2016). A very high proportion of young people use online (Azevedo et al., 2019), and online communication provides mutual support between survivors and is useful for cancer-related communication between young survivors and healthy peers (Hopkins et al., 2014; Maor and Mitchem, 2020; Yates, 2014). Thus, further research regarding the potential benefits and risks of online disclosure and effective disclosure strategies should help develop effective support for the self-disclosure of cancer-related experiences of young survivors.

4.3. Motivation for cancer disclosure and disclosure strategies and disadvantages of cancer disclosure

Overreaction by peers with little daily interaction and misinformation from peers about young survivors caused stress for the young survivors. Disadvantages of disclosure at the interpersonal level for young survivors included the risk of disclosing to individuals with whom they have unstable relationships, such as non-intimate classmates or work colleagues, similar to factors contributing to disadvantages of disclosure among adult survivors (Henderson et al., 2002; Munro et al., 2015). Young survivors also suppressed their own emotions as a strategy to protect the emotions of their loved ones. Notably, many parents of pediatric patients with cancer exhibit symptoms of post-traumatic stress disorder (Alderfer et al., 2005), and difficulty communicating about cancer is noted in their families (Ballard-Reisch and Letner, 2003; Pennant et al., 2019; Regan et al., 2015). This finding is consistent with the fact that adult survivors and their spouses avoid disclosing negative emotions to protect their partner (Porter et al., 2005). However, these strategies for adult survivors have been suggested to be associated with poorer mental health and reduced social support (Figueiredo et al., 2004; Langer et al., 2009). Similarly, emotional suppression in young survivors may have negative psychological effects on them.

Evidence of the effectiveness of interventions to support young survivors' cancer disclosure is limited. However, app-based interventions (Lazard et al., 2020) have shown potential to facilitate communication among young survivors. In addition, programs that promote psychological development related to reproductive health have reported increased cancer-related knowledge and reduced psychological distress and anxiety about romantic relationships (Canada et al., 2007). In the present study, young survivors had barriers to cancer disclosure to various targets (i.e. peers, friends, romantic partners, and family). Therefore, young survivors may benefit from the development of disclosure support programs, including disclosure skills for peers and friends.

Psychological support for the survivor, their family, and their partner is necessary to expand the benefits of disclosure and reduce the disadvantages for young survivors. The solutions to communication difficulties between survivors and their families related to cancer can be mutually beneficial (Bergeron et al., 2019; Feraco et al., 2018). Helping young survivors become more knowledgeable about cancer and develop communication skills will strengthen their disclosure strategies and make disclosure more profitable.

The impact on employment due to late complications and mental and physical limitations of young survivors (Kirchhoff et al., 2010; Stone et al., 2017) and difficulty obtaining insurance is a serious problem (Warner et al., 2016) related to survivors' financial difficulties. Preventing discrimination against young survivors requires an approach to thoroughly prohibit discrimination through legal compliance and to promote understanding of cancer experiences in schools, workplace, and society. Return-to-school support programs improve cancer awareness and positive attitudes toward survivors among teachers and peers (Helms et al., 2016). A supportive school or work culture promotes cancer disclosure (Stergiou-Kita et al., 2016). Additionally, a supportive environment without social constraints encourages acceptance of survivors' cancer-related thoughts and concerns to be disclosed as well as encourages survivors' own acceptance of their own cancer experiences (Lepore and Revenson, 2007).

4.4. Limitations and implications for research

There were several limitations to this study. First, the rigorous search strategy used in this study does not guarantee that all references on this topic were identified. As references were limited to articles written in English and Japanese, studies in all other languages were excluded. Second, the sample sizes of the studies included in this review were relatively small, and some studies included participants aged over 40

years. Participants' age at cancer onset also covered a wide range from pediatric to early adulthood cancers, and the types of cancer were also varied. The studies used substitute scales, since there is no scale specifically designed to evaluate cancer self-disclosure, and these limitations may have influenced the results of this study. Finally, the policy level, which does not directly affect disclosure, was not discussed in this review, and relationships between the self-disclosure and the policy level were not identified.

An analysis of the interactions among motivation, the social constraints, and disclosure strategies and their outcomes based on the SEM is necessary to expand the benefits of disclosure, reduce the disadvantages of disclosure for young cancer survivors, and remove disclosure barriers. To achieve this goal, more data need to be collected in multiple studies with larger sample sizes that focus on developmental stages and the social constraints.

5. Conclusions

Young survivors received benefits such as social support, self-empowerment, and self-control through SEM-level disclosure strategies that helped the disclosure recipient to gain a sense of relief, share a positive cancer story, and engage in advocacy. Young survivors' lack of knowledge about cancer, immature communication skills, lack of understanding of the disclosure recipient's cancer and family hurt from cancer disclosure were factors in inadequate communication. It resulted in their detriment, manifesting as frustration and the exposure of survivors' vulnerability. Other disadvantages, such as difficulty in finding employment and qualifying for insurance, require action at the political and institutional levels. Interventions to improve knowledge about cancer and develop disclosure skills as well as support for survivors' partners and families are necessary to ensure that disclosure by young survivors is beneficial to them. This review emphasizes the importance of developing a support program for the disclosure of cancer by young survivors and their families and for promoting knowledge and understanding of cancer in the general population.

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CRediT authorship contribution statement

Naoko Honda: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yui Shiroshita:** Writing – review & editing, Writing – original draft, Validation, Formal analysis. **Asuka Miyazaki:** Writing – review & editing, Validation, Formal analysis. **Ikuko Sobue:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis.

Declaration of competing interest

The authors declare no conflicts of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejon.2024.102542.

References

Adams, R.N., Winger, J.G., Mosher, C.E., 2015. A meta-analysis of the relationship between social constraints and distress in cancer patients. J. Behav. Med. 38, 294–305. https://doi.org/10.1007/s10865-014-9601-6.

Alderfer, M.A., Cnaan, A., Annunziato, R.A., Kazak, A.E., 2005. Patterns of posttraumatic stress symptoms in parents of childhood cancer survivors. J. Fam. Psychol. 19, 430–440. https://doi.org/10.1037/0893-3200.19.3.430.

Azevedo, R., Rosário, P., Martins, J., Rosendo, D., Fernández, P., Núñez, J.C., Magalhães, P., 2019. From the hospital bed to the laptop at home: effects of a blended self-regulated learning intervention. Int. J. Environ. Res. Publ. Health 16. https://doi.org/10.3390/ijerph16234802.

Ballard-Reisch, D.S., Letner, J.A., 2003. Centering families in cancer communication research: acknowledging the impact of support, culture and process on client/provider communication in cancer management. Patient Educ. Counsel. 50, 61–66. https://doi.org/10.1016/s0738-3991(03)00082-x.

Barnett, M., McDonnell, G., DeRosa, A., Schuler, T., Philip, E., Peterson, L., Touza, K., Jhanwar, S., Atkinson, T.M., Ford, J.S., 2016. Psychosocial outcomes and interventions among cancer survivors diagnosed during adolescence and young adulthood (AYA): a systematic review. J. Cancer Surviv. 10, 814–831. https://doi.org/10.1007/s11764-016-0527-6.

Belpame, N., Kars, M.C., Deslypere, E., Rober, P., Van Hecke, A., Verhaeghe, S., 2019. Living as a cancer survivor: a qualitative study on the experiences of Belgian adolescents and young adults (AYAs) after childhood cancer. J. Pediatr. Nurs. 49, e29–e35. https://doi.org/10.1016/j.pedn.2019.07.015.

Bergeron, S., Noskoff, K., Hayakawa, J., Frediani, J., 2019. Empowering adolescents and young adults to support, lead, and thrive: development and validation of an AYA oncology child life program. J. Pediatr. Nurs. 47, 1–6. https://doi.org/10.1016/j.pedn.2019.04.001.

Bleyer, A., 2007. Adolescent and young adult (AYA) oncology: the first A. Pediatr. Hematol. Oncol. 24, 325–336. https://doi.org/10.1080/08880010701316850.

Butner-Kozimor, L.M., Savla, J., 2020. Couples facing skin cancer in old age: a dyadic investigation of partner support and quality of life. Int. J. Aging Hum. Dev. 93, 746–766. https://doi.org/10.1177/0091415020940193.

Canada, A.L., Schover, L.R., Li, Y., 2007. A pilot intervention to enhance psychosexual development in adolescents and young adults with cancer. Pediatr. Blood Cancer 49, 824–828. https://doi.org/10.1002/pbc.21130.

Carpentier, M.Y., Fortenberry, J.D., Ott, M.A., Brames, M.J., Einhorn, L.H., 2011. Perceptions of masculinity and self-image in adolescent and young adult testicular cancer survivors: implications for romantic and sexual relationships. Psycho Oncol. 20, 738–745. https://doi.org/10.1002/pon.1772.

Cassano, J., Nagel, K., O'Mara, L., 2008. Talking with others who "just know": perceptions of adolescents with cancer who participate in a teen group. J. Pediatr. Oncol. Nurs. 25, 193–199. https://doi.org/10.1177/1043454208319972.

Chaudoir, S.R., Fisher, J.D., 2010. The disclosure processes model: understanding disclosure decision making and postdisclosure outcomes among people living with a concealable stigmatized identity. Psychol. Bull. 136, 236–256. https://doi.org/10.1037/a0018193.

Clarke, V., Braun, V., 2014. Thematic analysis. In: Teo, T. (Ed.), Encyclopedia of Critical Psychology. Springer, New York,NY. https://doi.org/10.1007/978-1-4614-5583-7_311.

Close, A.G., Dreyzin, A., Miller, K.D., Seynnaeve, B.K.N., Rapkin, L.B., 2019. Adolescent and young adult oncology—past, present, and future. CA A Cancer J. Clin. 69, 485–496. https://doi.org/10.3322/caac.21585.

Dang, D., Dearholt, S.L., 2017. Johns Hopkins Nursing Evidence-Based Practice, third ed. Model and Guidelines. Sigma Theta Tau International, Indianapolis.

Darabos, K., Hoyt, M.A., Ford, J.S., 2019. Relational processes and psychological adjustment among young adults with cancer: the role of technology-related communication. Psycho Oncol. 28, 2406–2414. https://doi.org/10.1002/pon.5269.

Desandes, E., Stark, D.P., 2016. Epidemiology of adolescents and young adults with cancer in europe. Prog Tumor Res 43, 1–15. https://doi.org/10.1159/000447037.

Dienlin, T., Metzger, M.J., 2016. An extended privacy calculus model for SNSs: analyzing self-disclosure and self-withdrawal in a representative U.S. Sample. J Comput-Mediat Comm 21, 368–383. https://doi.org/10.1111/jcc4.12163.

Dobinson, K.A., Hoyt, M.A., Seidler, Z.E., Beaumont, A.L., Hullmann, S.E., Lawsin, C.R., 2016. A grounded theory investigation into the psychosexual unmet needs of adolescent and young adult cancer survivors. J. Adolesc. Young Adult Oncol. 5, 135–145. https://doi.org/10.1089/jayao.2015.0022.

Easley, J., 2019. Motivations for cancer history disclosure among young adult cancer survivors. J. Cancer Surviv. 13, 447–458. https://doi.org/10.1007/s11764-019-00766-w.

Eddy, E.M., 1981. The ecology of human-development - bronfenbrenner. U. American Anthropologist 83. https://doi.org/10.1525/aa.1981.83.3.02a00220, 643-643.

Else-Quest, N.M., Jackson, T.L., 2014. Cancer Stigma, the Stigma of Disease and Disability: Understanding Causes and Overcoming Injustices. American Psychological Association, Washington, DC, US, pp. 165–181. https://doi.org/10.1037/14297-009.

Emetu, R.E., Rivera, G., 2018. After sexual identity disclosure: an ecological perceptive of LGB young adults. Am. J. Health Behav. 42, 45–60. https://doi.org/10.5993/AJHB.42.4.5.

Feraco, A.M., Brand, S.R., Gagne, J., Sullivan, A., Block, S.D., Wolfe, J., 2018. Development of the "Day 100 Talk": addressing existing communication gaps during the early cancer treatment period in childhood cancer. Pediatr. Blood Cancer 65, e26972. https://doi.org/10.1002/pbc.26972.

Figueiredo, M.I., Fries, E., Ingram, K.M., 2004. The role of disclosure patterns and unsupportive social interactions in the well-being of breast cancer patients. Psycho Oncol. 13, 96–105. https://doi.org/10.1002/pon.717.

Fujisawa, D., Umezawa, S., Fujimori, M., Miyashita, M., 2020. Prevalence and associated factors of perceived cancer-related stigma in Japanese cancer survivors. Jpn. J. Clin. Oncol. 50, 1325–1329. https://doi.org/10.1093/jjco/hyaa135.

Garrard, J., 2013. Health Sciences Literature Review Made Easy: the Matrix Method. Jones & Bartlett Learning, Massachusetts.

Greene, K., Magsamen-Conrad, K., Venetis, M.K., Checton, M.G., Bagdasarov, Z., Banerjee, S.C., 2012. Assessing health diagnosis disclosure decisions in relationships: testing the disclosure decision-making model. Health Commun. 27, 356–368. https://doi.org/10.1080/10410236.2011.586988.

Grunfeld, E., Cooper, A., 2012. A longitudinal qualitative study of the experience of working following treatment for gynaecological cancer. Psycho Oncol. 21, 82–89. https://doi.org/10.1002/pon.1874.

Hagedoorn, M., Kuijer, R.G., Buunk, B.P., DeJong, G.M., Wobbes, T., Sanderman, R., 2000. Marital satisfaction in patients with cancer: does support from intimate partners benefit those who need it most? Health Psychol. 19, 274. https://doi.org/10.1037//0278-6133.19.3.274.

Hasson-Ohayon, I., Goldzweig, G., Braun, M., Hagedoorn, M., 2022. Beyond "being open about it": a systematic review on cancer related communication within couples. Clin. Psychol. Rev. 96, 102176 https://doi.org/10.1016/j.cpr.2022.102176.

Helms, A.S., Schmiegelow, K., Brok, J., Johansen, C., Thorsteinsson, T., Simovska, V., Larsen, H.B., 2016. Facilitation of school re-entry and peer acceptance of children with cancer: a review and meta-analysis of intervention studies. Eur. J. Cancer Care 25, 170–179. https://doi.org/10.1111/ecc.12230.

Henderson, B.N., Davison, K.P., Pennebaker, J.W., Gatchel, R.J., Baum, A., 2002. Disease disclosure patterns among breast cancer patients. Psychol. Health 17, 51–62. https://doi.org/10.1080/08870440290001520.

Hopia, H., Latvala, E., Liimatainen, L., 2016. Reviewing the methodology of an integrative review. Scand. J. Caring Sci. 30, 662–669. https://doi.org/10.1111/scs.12327.

Hopkins, L., Moss, J., Green, J., Strong, G., 2014. Embedding learning in a paediatric hospital: changing practice and keeping connected. Int. J. Incl. Educ. 18, 312–321. https://doi.org/10.1080/13603116.2012.752877.

Hughes, N., Stark, D., 2018. The management of adolescents and young adults with cancer. Cancer Treat Rev. 67, 45–53. https://doi.org/10.1016/j.ctrv.2018.05.001.

Irene Su, H., Lee, Y.T., Barr, R., 2018. Oncofertility: meeting the fertility goals of adolescents and young adults with cancer. Cancer J. 24, 328–335. https://doi.org/10.1097/ppo.0000000000000344.

Janin, M.M.H., Ellis, S.J., Wakefield, C.E., Fardell, J.E., 2018. Talking about cancer among adolescent and young adult cancer patients and survivors: a systematic review. J. Adolesc. Young Adult Oncol. 7, 515–524. https://doi.org/10.1089/jayao.2017.0131.

Kaplan, S., Garrick, B.J., 1981. On the quantitative definition of risk. Risk Anal. 1, 11–27. https://doi.org/10.1111/j.1539-6924.1981.tb01350.x.

Ke, Y., Tan, C.J., Ng, T., Tan, I.M.J., Goh, W.L., Poon, E., Farid, M., Neo, P.S.H., Srilatha, B., Chan, A., 2020. Optimizing survivorship care services for asian adolescent and young adult cancer survivors: a qualitative study. J. Adolesc. Young Adult Oncol. 9, 384–393. https://doi.org/10.1089/jayao.2019.0145.

Kim, M.A., Yi, J., 2014. Life after cancer: how does public stigma increase psychological distress of childhood cancer survivors? Int. J. Nurs. Stud. 51, 1605–1614. https://doi.org/10.1016/j.ijnurstu.2014.04.00505-1614.

Kim, M.A., Yi, J., Prince, K.C., Nagelhout, E., Wu, Y.P., 2018. The effects of individual- and network-level factors on discussion of cancer experiences: survivors of childhood cancer in Korea. J. Psychosoc. Oncol. 36, 31–48. https://doi.org/10.1080/07347332.2017.1379582.

Kirchhoff, A.C., Leisenring, W., Krull, K.R., Ness, K.K., Friedman, D.L., Armstrong, G.T., Stovall, M., Park, E.R., Oeffinger, K.C., Hudson, M.M., Robison, L.L., Wickizer, T., 2010. Unemployment among adult survivors of childhood cancer: a report from the childhood cancer survivor study. Med. Care 48, 1015–1025. https://doi.org/10.1097/mlr.0b013e3181eaf880.

Krasnova, H., Spiekermann, S., Koroleva, K., Hildebrand, T., 2010. Online social networks: why we disclose. J. Inf. Technol. 25, 109–125. https://doi.org/10.1057/jit.2010.6.

Lahaye, M., Aujoulat, I., Vermylen, C., Brichard, B., 2017. Long-term effects of haematopoietic stem cell transplantation after pediatric cancer: a qualitative analysis of life experiences and adaptation strategies. Front. Psychol. 8, 704. https://doi.org/10.3389/fpsyg.2017.00704.

Langer, S.L., Brown, J.D., Syrjala, K.L., 2009. Intrapersonal and interpersonal consequences of protective buffering among cancer patients and caregivers. Cancer 115, 4311–4325. https://doi.org/10.1002/cncr.24586.

Lazard, A.J., Saffer, A.J., Horrell, L., Benedict, C., Love, B., 2020. Peer-to-peer connections: perceptions of a social support app designed for young adults with cancer. Psycho Oncol. 29, 173–181. https://doi.org/10.1002/pon.5220.

Lepore, S.J., 2001. A social–cognitive processing model of emotional adjustment to cancer. In: Baum, A., Andersen, B.L. (Eds.), Psychosocial Interventions for Cancer. American Psychological Association, Washington, DC, pp. 99–116. https://doi.org/10.1037/10402-006.

Lepore, S.J., Revenson, T.A., 2007. Social constraints on disclosure and adjustment to cancer. Soc. Personal. Psychol. Compass 1, 313–333. https://doi.org/10.1111/j.1751-9004.2007.00013.x.

Lepore, S.J., Wortman, C.B., Silver, R.C., Wayment, H.A., 1996. Social constraints, intrusive thoughts, and depressive symptoms among bereaved mothers. J. Pers. Soc. Psychol. 70, 271–282. https://doi.org/10.1037//0022-3514.70.2.271.

Lyons, K.S., Miller, L.M., McCarthy, M.J., 2016. The roles of dyadic appraisal and coping in couples with lung cancer. J. Fam. Nurs. 22, 493–514. https://doi.org/10.1177/1074840716675976.

Manne, S.L., Kissane, D., Zaider, T., Kashy, D., Lee, D., Heckman, C., Virtue, S.M., 2015. Holding back, intimacy, and psychological and relationship outcomes among couples coping with prostate cancer. J. Fam. Psychol. 29, 708–719. https://doi.org/10.1037/fam0000096.

Maor, D., Mitchem, K., 2020. Hospitalized adolescents' use of mobile technologies for learning, communication, and well-being. J. Adolesc. Res. 35, 225–247. https://doi.org/10.1177/0743558417753953.

Marzorati, C., Riva, S., Pravettoni, G., 2017. Who is a cancer survivor? A systematic review of published definitions. J. Cancer Educ. 32, 228–237. https://doi.org/10.1007/s13187-016-0997-2.

McLeroy, K.R., Bibeau, D., Steckler, A., Glanz, K., 1988. An ecological perspective on health promotion programs. Health Educ. Q. 15, 351–377. https://doi.org/10.1177/109019818801500401.

Meidlinger, P.C., Hope, D.A., 2014. Differentiating disclosure and concealment in measurement of outness for sexual minorities: the Nebraska outness scale. Psychol Sex Orientat Gend Divers 1, 489–497. https://doi.org/10.1037/sgd0000080.

Miller, K.D., Fidler-Benaoudia, M., Keegan, T.H., Hipp, H.S., Jemal, A., Siegel, R.L., 2020. Cancer statistics for adolescents and young adults, 2020. CA A Cancer J. Clin. 70, 443–459. https://doi.org/10.3322/caac.21637.

Moore, A.R., Buchanan, N.D., Fairley, T.L., Lee Smith, J., 2015. Public health action model for cancer survivorship. Am. J. Prev. Med. 49, S470–S476. https://doi.org/10.1016/j.amepre.2015.09.001.

Munro, H., Scott, S.E., King, A., Grunfeld, E.A., 2015. Patterns and predictors of disclosure of a diagnosis of cancer. Psycho Oncol. 24, 508–514. https://doi.org/10.1002/pon.3679.

Partelow, S., 2018. A review of the social-ecological systems framework: applications, methods, modifications, and challenges. Ecol. Soc. 23 https://doi.org/10.5751/es-10594-230436.

Pennant, S., C Lee, S., Holm, S., Triplett, K.N., Howe-Martin, L., Campbell, R., Germann, J., 2019. The role of social support in adolescent/young adults coping with cancer treatment. Children 7. https://doi.org/10.3390/children7010002.

Perales, M.A., Drake, E.K., Pemmaraju, N., Wood, W.A., 2016. Social media and the adolescent and young adult (AYA) patient with cancer. Curr. Hematol. Malig. Rep. 11, 449–455. https://doi.org/10.1007/s11899-016-0313-6.

Pini, S., Hugh-Jones, S., Shearsmith, L., Gardner, P., 2019. 'What are you crying for? I don't even know you' – the experiences of teenagers communicating with their peers when returning to school. Eur. J. Oncol. Nurs. 39, 28–34. https://doi.org/10.1016/j.ejon.2018.12.010.

Porter, L.S., Keefe, F.J., Hurwitz, H., Faber, M., 2005. Disclosure between patients with gastrointestinal cancer and their spouses. Psycho Oncol. 14, 1030–1042. https://doi.org/10.1002/pon.915.

Qiao, S., Li, X., Zhou, Y., Shen, Z., Tang, Z., Stanton, B., 2015. Factors influencing the decision-making of parental HIV disclosure: a socio-ecological approach. AIDS 29 (Suppl. 1), S25–S34. https://doi.org/10.1097/qad.0000000000000670.

Rabin, C., 2019. Self-disclosure to peers by young adult cancer survivors. Psycho Oncol. 28, 181–186. https://doi.org/10.1002/pon.4930.

Rabin, C., 2020. Cancer-related self-disclosure in the workplace/school by adolescent and young adult cancer survivors. J. Adolesc. Young Adult Oncol. 9, 528–533. https://doi.org/10.1089/jayao.2019.0159.

Regan, T.W., Lambert, S.D., Kelly, B., Falconier, M., Kissane, D., Levesque, J.V., 2015. Couples coping with cancer: exploration of theoretical frameworks from dyadic studies. Psycho Oncol. 24, 1605–1617. https://doi.org/10.1002/pon.3854.

Restall, G., Diaz, F., Faucher, P., Roger, K., 2021. Participatory design and qualitative evaluation of a decision guide for workplace human immunodeficiency virus self-disclosure: the importance of a socio-ecological perspective. Health Expect. 24, 1220–1229. https://doi.org/10.1111/hex.13252.

Saloustros, E., Stark, D.P., Michailidou, K., Mountzios, G., Brugieres, L., Peccatori, F.A., Jezdic, S., Essiaf, S., Douillard, J.-Y., Bielack, S., 2017. The care of adolescents and young adults with cancer: results of the ESMO/SIOPE survey. ESMO Open 2, e000252. https://doi.org/10.1136/esmoopen-2017-000252.

Schilstra, C.E., Fardell, J.E., Burns, M.A., Ellis, S.J., Anazodo, A.C., Trahair, T.N., Sansom-Daly, U.M., 2021. Determinants of social functioning among adolescents and young adults with cancer: a systematic review. Psycho Oncol. 30, 1626–1642. https://doi.org/10.1002/pon.5740.

Stergiou-Kita, M., Pritlove, C., Kirsh, B., 2016. The "Big C"-stigma, cancer, and workplace discrimination. J. Cancer Surviv. 10, 1035–1050. https://doi.org/10.1007/s11764-016-0547-2.

Stone, D.S., Ganz, P.A., Pavlish, C., Robbins, W.A., 2017. Young adult cancer survivors and work: a systematic review. J. Cancer Surviv. 11, 765–781. https://doi.org/10.1007/s11764-017-0614-3.

Thompson, A.L., Long, K.A., Marsland, A.L., 2013. Impact of childhood cancer on emerging adult survivors' romantic relationships: a qualitative account. J. Sex. Med. 10 (Suppl. 1), 65–73. https://doi.org/10.1111/j.1743-6109.2012.02950.x.

Tsuchiya, M., Adachi, K., Kumagai, K., Kondo, N., Kimata, A., 2022. Cancer disclosure to friends: survey on psychological distress and perceived social support provision. Eur. J. Cancer Care 31, e13332. https://doi.org/10.1111/ecc.13332.

Tsuchiya, M., Horn, S., Ingham, R., 2015. Social disclosure about lymphoedema symptoms: a qualitative study among Japanese breast cancer survivors. Psychol. Health Med. 20, 680–684. https://doi.org/10.1080/13548506.2014.986140.

Warner, E.L., Kent, E.E., Trevino, K.M., Parsons, H.M., Zebrack, B.J., Kirchhoff, A.C., 2016. Social well-being among adolescents and young adults with cancer: a systematic review. Cancer 122, 1029–1037. https://doi.org/10.1002/cncr.29866.

Wertheim, R., Goldzweig, G., Mashiach-Eizenberg, M., Pizem, N., Shacham-Shmueli, E., Hasson-Ohayon, I., 2018. Correlates of concealment behavior among couples coping with cancer: actor partner model. Psycho Oncol. 27, 583–589. https://doi.org/10.1002/pon.4552.

Whittemore, R., Knafl, K., 2005. The integrative review: updated methodology. J. Adv. Nurs. 52, 546–553. https://doi.org/10.1111/j.1365-2648.2005.03621.x.

Wong, A.W.K., Chang, T.T., Christopher, K., Lau, S.C.L., Beaupin, L.K., Love, B., Lipsey, K.L., Feuerstein, M., 2017. Patterns of unmet needs in adolescent and young adult (AYA) cancer survivors: in their own words. J. Cancer Surviv. 11, 751–764. https://doi.org/10.1007/s11764-017-0613-4.

World Health Organization. Adolescent health. https://www.who.int/health-topics/adolescent-health/#tab=tab_1 (accessed 13 April 2023).

Yates, L., 2014. Young people with health conditions and the inclusive education problematic. Int. J. Incl. Educ. 18, 283–294. https://doi.org/10.1080/13603116.2012.700737.

Yi, H.J., Nam, S.I., 2017. The effect of advocacy for overcoming stigma on posttraumatic growth: focusing on childhood cancer survivors. Soc. Work. Health Care 56, 840–854. https://doi.org/10.1080/00981389.2017.1353569.

Yoo, G.J., Aviv, C., Levine, E.G., Ewing, C., Au, A., 2010. Emotion work: disclosing cancer. Support. Care Cancer 18, 205–215. https://doi.org/10.1007/s00520-009-0646-y.

Zebrack, B., Isaacson, S., 2012. Psychosocial care of adolescent and young adult patients with cancer and survivors. J. Clin. Oncol. 30, 1221–1226. https://doi.org/10.1200/jco.2011.39.5467.

Zhang, M.M., Yang, Y.J., Su, D., Zhang, T., Jiang, X.X., Li, H.P., 2019. A randomized controlled trial of a guided self-disclosure intervention to facilitate benefit finding in Chinese breast cancer patients: study protocol. J. Adv. Nurs. 75, 1805–1814. https://doi.org/10.1111/jan.14042.